

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA14

COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO4: *Examine intermediate code generation techniques to enhance code optimization (BL4)*

Assessment Tool Weightage: 40%

QUESTIONS: 4

Total Marks:40

Annexure A

CLASS TEST

Code of Conduct:

I Monika P (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Monika P

Q. No	Understanding of Concepts (25)	Explanation (25)	Application (20)	Organization(15)	Accuracy(10)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 30	/ 20	/ 15	/ 10	/ 100

Annexure A

CLASS TEST

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	15%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps

1. Three Address Code for the if-else statement

if $((a < b) \text{ and } (c < d) \text{ or } (a > d))$ then

$z = x + y * z$

else

$z = z + 1$

38/40

Step 1 :- Understand the Boolean condition

$(a < b) \text{ AND } (c < d) \text{ OR } (a > d)$

The three relations are: $a < b$, $c < d$, $a > b$.

Step 2 :- Create labels

Use labels for the condition checks, then
part else part, and the end of the
statement. $L_1, L_2, L_3, L_4, L_5, L_6$.

Step 3 :- Generate TAC for the condition.

L_1 :

if $a < b$ goto L_2

goto L_5

L2:
if $c < d$ goto L4

goto L3

L3:

if $a > b$ goto L4

goto L5

Step 4: Generate TAC for the THEN Part

L4:

$t_1 = y * z$

$t_2 = x + t_1$

$z = t_2$

goto L6

Step 5: Generate TAC for the ELSE part

L5:

$t_3 = z + 1$

$z = t_3$

~~Figure~~

2. Three Address code Using Backpatching

Boolean expression $(a < b)$ and $(c < d)$ or $(e < f)$

Step 1:- Apply Operator precedence

$[(a < b) \text{ AND } (c < d)] \text{ or } (e < f)$

Step 2:- Generate incomplete jumps

1. if $a < b$ goto -

2. goto -

3. if $c < d$ goto -

4. goto -

5. if $e < f$ goto -

6. goto -

Step 3:- Handle AND

for $(a < b)$ AND $(c < d)$, if $a < b$ is true, control must continue to the second condition.

Backpatch statement 1 to statement 3.

1. if $a < b$ goto 3.

Step 4: Handle OR

Backpatch statement 2 to statement 5.

Backpatch statement 4 to statement 5.

Step 5: final Backpatch code

1. if $a < b$ goto 3

2. goto 5

3. if $c < d$ goto 7

4. goto 5

5. if $e < f$ goto 7

6. goto 8

7. TRUE

8. FALSE

4. TAC for Boolean Expression Using the Expression $(a > b) \text{ or } (c < d) \text{ or } (e < f)$

Step 1 :- Understand the expression

$[(a > b) \text{ or } (c < d)] \text{ or } (e < f)$

Step 2 :- Check the first condition

L1 :

if $a > b$ goto LTRUE

goto L3

Step 3 :- Check the second condition

L2 :

if $c < d$ goto LTRUE

goto L3

Step 4 : check the third condition

L3 :

if $e < f$ goto LTRUE

goto LFALSE

Final TAC for Q4

L1 :

if $a > b$ goto LTRUE

goto L2

L2 :

if $c < d$ goto LTRUE

goto L2

L3:

if $c < f$ goto LTRUE

goto LFALSE

LTRUE:

// Boolean expression is true
goto LEND

LFALSE:

// Boolean expression is false

LEND:

5. TAC for the while-if-else Program

while ($A < C$ and $B > D$) do

if $A = 1$ then

$C = C + 1$

else

while $A \leq D$ do

$A = A + B$

Step 1 :- Construct the Outer while condition

L1:

if $A < C$ goto L2

goto LEND

L2:

if $B > D$ goto L3

goto LEND

Step 2: Construct the If condition

L3:

if $A == 1$ goto L4

goto L5

L4 is the THEN statement

Step 4:- Construct the ELSE block and inner while.

L5:

if $A < = D$ goto L6

goto L1

Step 5: Construct the inner while body

L6:

$t_2 = A + B$

$A = t_2$

goto L5

Complete TAC for Q5

L1:

if $A < C$ goto L2

goto LEND

L2:

if $B > D$ goto L3

goto LEND

L3:

if $A == 1$ goto L4

goto L5

L4:

$t_1 = C + 1$

$C = t_1$

goto L1

L5:

if $A < = D$ goto L6

goto L1

L6:

$t_2 = A + B$

$A = t_2$

goto L5

LEND: